

Python Coding Assignment

Lists · Dictionaries · Loops · Conditions · Comprehensions

Topics	Lists · Dictionaries · for loop · if / elif / else · break · continue · List comprehension · Dict comprehension
Difficulty	6 Basic · 8 Easy · 10 Intermediate · 6 Medium
Total	30 questions

Section — Basic Q1 – Q6

Single concept per question. Focus on getting the syntax right before combining ideas.

Q1 Print all items in a list	
BASIC	Loop
Given a list of fruits, use a for loop to print each fruit on a new line.	
INPUT / SETUP <pre>fruits = ["apple", "banana", "cherry", "mango"]</pre>	EXPECTED OUTPUT <pre>apple banana cherry mango</pre>

Q2 Sum all numbers in a list	
BASIC	Loop
Given a list of numbers, use a for loop to calculate and print their total sum. Do not use the built-in sum() function.	
INPUT / SETUP <pre>nums = [4, 7, 2, 9, 1, 5]</pre>	EXPECTED OUTPUT <pre>28</pre>

Q3 Count even numbers in a list

BASIC

Loop

Given a list of integers, use a for loop and an if statement to count how many numbers are even.

INPUT / SETUP

```
nums = [1, 2, 3, 4, 5, 6, 7, 8]
```

EXPECTED OUTPUT

```
4
```

Q4 Print all keys and values from a dictionary

BASIC

Dictionary

Given a student dictionary, use a for loop to print each key and its value on one line.

INPUT / SETUP

```
student = {"name": "Ali", "age": 20,  
"grade": "A", "city": "Toronto"}
```

EXPECTED OUTPUT

```
name : Ali  
age : 20  
grade : A  
city : Toronto
```

Q5 Find the largest number in a list

BASIC

Loop

Without using max(), loop through a list and track the largest number seen so far.

INPUT / SETUP

```
nums = [3, 7, 1, 9, 4, 6, 2]
```

EXPECTED OUTPUT

```
9
```

Q6 Skip a specific item using continue

BASIC	Loop
Loop through a list of names and print each one — but skip the name 'Bob' using continue.	
INPUT / SETUP <pre>names = ["Alice", "Bob", "Charlie", "Diana", "Bob", "Eve"]</pre>	EXPECTED OUTPUT <pre>Alice Charlie Diana Eve</pre>

Section — Easy Q7 – Q14

One or two concepts combined. Introduces break, building new lists inside loops, and looping over dictionaries.

Q7 Stop a loop early using break	
EASY	Loop
Loop through a list of numbers and print each one. As soon as you encounter the number 0, stop the loop using break. Do not print 0.	
INPUT / SETUP <pre>nums = [5, 3, 8, 0, 2, 7, 1]</pre>	EXPECTED OUTPUT <pre>5 3 8</pre>

Q8 Separate odd and even numbers into two lists	
EASY	List
Given a list of integers, use a for loop and if/else to build two separate lists — one for even numbers and one for odd numbers.	
INPUT / SETUP <pre>nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]</pre>	EXPECTED OUTPUT <pre>Evens: [2, 4, 6, 8, 10] Odds: [1, 3, 5, 7, 9]</pre>

Q9 Count how many students passed

EASY

Dictionary

Given a dictionary of student names and scores, loop through it and count how many students scored 50 or above (passing mark).

INPUT / SETUP

```
scores = {"Ali":72, "Mia":45,  
"Joe":88, "Sam":39, "Eva":60,  
"Leo":50}
```

EXPECTED OUTPUT

```
4 students passed
```

Q10 Reverse a list using a loop

EASY

List

Without using `.reverse()` or slicing, use a for loop to build a new reversed version of the list.

INPUT / SETUP

```
lst = [1, 2, 3, 4, 5]
```

EXPECTED OUTPUT

```
[5, 4, 3, 2, 1]
```

Q11 Print a multiplication table

EASY

Loop

Given a number `n`, use a for loop to print its multiplication table from 1 to 10.

INPUT / SETUP

```
n = 7
```

EXPECTED OUTPUT

```
7 x 1 = 7  
7 x 2 = 14  
...  
7 x 10 = 70
```

Q12 Find the student with the highest score

EASY

Dictionary

Given a dictionary of student names and scores, loop through it to find and print the name of the student with the highest score.

INPUT / SETUP

```
scores = {"Ali":72, "Mia":95,  
"Joe":88, "Sam":61, "Eva":90}
```

EXPECTED OUTPUT

```
Mia
```

Q13 Remove all negative numbers from a list

EASY

List

Given a list of integers, use a for loop and an if statement to build a new list containing only non-negative numbers.

INPUT / SETUP

```
nums = [3, -1, 4, -7, 5, -2, 8, 0]
```

EXPECTED OUTPUT

```
[3, 4, 5, 8, 0]
```

Q14 Find the first repeated item in a list

EASY

Loop

Loop through a list and track seen items. Use break to stop as soon as you find the first item that has appeared before.

INPUT / SETUP

```
items = ["apple", "banana", "cherry",  
"banana", "mango", "apple"]
```

EXPECTED OUTPUT

```
First repeated item: banana
```

Section — Intermediate Q15 – Q24

Multiple concepts combined. Includes list and dict comprehensions, FizzBuzz, grouping, and using break and continue together.

Q15 Squares of even numbers using list comprehension

INTERMEDIATE

Comprehension

Use a single list comprehension to produce a list of squares of all even numbers from 1 to 20.

INPUT / SETUP

```
# Work with range(1, 21)
```

EXPECTED OUTPUT

```
[4, 16, 36, 64, 100, 144, 196, 256, 324, 400]
```

Q16 Filter long words using list comprehension

INTERMEDIATE

Comprehension

Given a list of words, use a list comprehension to keep only words that are longer than 4 characters.

INPUT / SETUP

```
words =  
["cat", "elephant", "dog", "python", "ox",  
"tiger", "ant", "crocodile"]
```

EXPECTED OUTPUT

```
["elephant", "python", "tiger",  
"crocodile"]
```

Q17 Invert a dictionary using a for loop

INTERMEDIATE

Dictionary

Given a dictionary, loop through it and build a new dictionary where keys and values are swapped.

INPUT / SETUP

```
codes = {"IN": "India", "US": "USA",  
"CA": "Canada", "UK": "Britain"}
```

EXPECTED OUTPUT

```
{"India": "IN", "USA": "US",  
"Canada": "CA", "Britain": "UK"}
```

Q18 Group numbers into positive, negative, and zero

INTERMEDIATE

Mixed

Given a list of numbers, loop through and sort them into three separate lists using if/elif/else.

INPUT / SETUP

```
nums = [4, -2, 0, 7, -5, 0, 3, -1, 8]
```

EXPECTED OUTPUT

```
Positive: [4, 7, 3, 8]
Negative: [-2, -5, -1]
Zero:     [0, 0]
```

Q19 Build a dict comprehension of word lengths

INTERMEDIATE

Comprehension

Given a list of words, use a dict comprehension to create a dictionary where each word is the key and its length is the value.

INPUT / SETUP

```
words = ["python", "loop", "dict",
         "comprehension", "list"]
```

EXPECTED OUTPUT

```
{"python":6, "loop":4, "dict":4,
 "comprehension":13, "list":4}
```

Q20 FizzBuzz

INTERMEDIATE

Mixed

Loop through numbers 1 to 30. Print 'Fizz' for multiples of 3, 'Buzz' for multiples of 5, 'FizzBuzz' for multiples of both, and the number otherwise.

INPUT / SETUP

```
# Work with range(1, 31)
```

EXPECTED OUTPUT

```
1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11
Fizz 13 14 FizzBuzz 16 17 Fizz 19
Buzz ...
```

Q21 Find all common elements between two lists

INTERMEDIATE

List

Given two lists, use a for loop and an if statement to find all elements that appear in both lists. Store results in a new list.

INPUT / SETUP

```
a = [1, 2, 3, 4, 5, 6]
b = [4, 5, 6, 7, 8, 9]
```

EXPECTED OUTPUT

```
[4, 5, 6]
```

Q22 Count word frequency in a sentence

INTERMEDIATE

Dictionary

Given a sentence, split it into words and use a for loop with if/else to build a dictionary that counts how many times each word appears.

INPUT / SETUP

```
sentence = "the cat sat on the mat
the cat"
```

EXPECTED OUTPUT

```
{"the":3, "cat":2, "sat":1, "on":1,
"mat":1}
```

Q23 Skip multiples of 3 and stop at 50

INTERMEDIATE

Loop

Loop through numbers 1 to 100. Use continue to skip any multiple of 3, and use break to stop once you exceed 50. Print every number that passes through.

INPUT / SETUP

```
# Work with range(1, 101)
```

EXPECTED OUTPUT

```
1 2 4 5 7 8 10 11 ... 49 50
```

Q24 Flatten a nested list using list comprehension

INTERMEDIATE	Comprehension
Given a list of lists, use a list comprehension with two for clauses to flatten it into a single list.	
INPUT / SETUP <pre>nested = [[1, 2, 3], [4, 5], [6, 7, 8, 9]]</pre>	EXPECTED OUTPUT <pre>[1, 2, 3, 4, 5, 6, 7, 8, 9]</pre>

Section — Medium Q25 – Q30

Real problem-solving. Requires combining loops, conditionals, lists, and dictionaries — or using comprehensions with conditions.

Q25 Build a frequency dictionary and find the most common item	
MEDIUM	Mixed
Given a list of items, use a for loop to build a frequency dictionary. Then loop through the dictionary to find the item that appears the most.	
INPUT / SETUP <pre>items = ["apple", "banana", "apple", "cherry", "banana", "apple", "mango", "banana"]</pre>	EXPECTED OUTPUT <pre>apple: 3 banana: 3 cherry: 1 mango: 1 Most common: apple</pre>

Q26 Dict comprehension — pass or fail per student	
MEDIUM	Comprehension
Given a dictionary of student scores, use a dict comprehension to create a new dictionary where each student maps to 'Pass' if score $\geq$ 50, otherwise 'Fail'.	
INPUT / SETUP <pre>scores = {"Ali":72, "Mia":45, "Joe":88, "Sam":39, "Eva":60}</pre>	EXPECTED OUTPUT <pre>{"Ali":"Pass", "Mia":"Fail", "Joe":"Pass", "Sam":"Fail", "Eva":"Pass"}</pre>

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Q27 Find the second largest number without sorting	
MEDIUM	Mixed
Loop through a list using if/elif to track both the largest and second largest number as you go. Do not sort the list.	
INPUT / SETUP <pre>nums = [3, 7, 1, 9, 4, 6, 8, 2]</pre>	EXPECTED OUTPUT <pre>Second largest: 8</pre>

Q28 Group words by their first letter	
MEDIUM	Mixed
Given a list of words, use a for loop and if/else to build a dictionary where each key is a letter and the value is a list of words starting with that letter.	
INPUT / SETUP <pre>words = ["apple", "avocado", "banana", "blueberry", "cherry", "apricot", "cantaloupe"]</pre>	EXPECTED OUTPUT <pre>{"a": ["apple", "avocado", "apricot"], "b": ["banana", "blueberry"], "c": ["cherry", "cantaloupe"]}</pre>

Q29 List comprehension with nested condition	
MEDIUM	Comprehension
Use a list comprehension to go through numbers 1 to 50 and keep only those that are divisible by 2 but NOT divisible by 6.	
INPUT / SETUP <pre># Work with range(1, 51)</pre>	EXPECTED OUTPUT <pre>[2, 4, 8, 10, 14, 16, 20, 22, 26, 28, 32, 34, 38, 40, 44, 46, 50]</pre>

Q30 Rotate a list by n positions using a loop

MEDIUM

Mixed

Given a list and a number n, use a for loop to move the first n items to the end of the list one at a time using list operations. Do not use slicing.

INPUT / SETUP

```
lst = [1, 2, 3, 4, 5, 6, 7]  
n = 3
```

EXPECTED OUTPUT

```
[4, 5, 6, 7, 1, 2, 3]
```

Python Coding Assignment · Good luck!